Structural Vibration of Axial-Flux and Radial-Flux Permanent Magnet Synchronous Motors for Electric Vehicle Traction: A Finite-Element Comparison

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***Abstract*—Structural vibration is a key concern in electric vehicle traction motors, and radial-flux and axial-flux topologies apply their electromagnetic forces along different directions. The two architectures have been compared for efficiency and torque density but not, under controlled conditions, for structural vibration. This study compares finite-element models of a radial-flux stator and an axial-flux rotor disc, the first compliant member in each machine's force path, under matched mass, pole count, and current density. At rated speed and the 400 Hz axial harmonic, the disc's out-of-plane displacement exceeds the stator's radial displacement by a factor that ranges from about 86 to 2.6×10^4 depending on how each part is supported, with an intermediate support pair giving about 6×10^3. A decomposition attributes the gap mainly to structural compliance, which combines support condition, component geometry, and the spatial order of the load, rather than to force magnitude (about 2.2x) or dynamic amplification (about 1.3x). The gap narrows with stiffer supports but does not close in any configuration tested. A finite-element speed sweep further shows that the fourth electrical harmonic crosses the disc's umbrella mode at 3,285 r/min, only 9.5% above rated speed, and that at rated speed this 800 Hz harmonic drives a disc response of about 126 μm, roughly 3.5 times the 400 Hz response. Axial-flux vibration is therefore governed first by rotor structural design rather than by electromagnetic force magnitude.**

***Index Terms*—Axial-flux permanent magnet (AFPM) motors, finite-element analysis (FEA), radial-flux permanent magnet (RFPM) motors, structural stiffness, structural vibration, vibration decomposition.**

I. Introduction

ELECTRIC vehicle (EV) traction motors generate high-frequency electromagnetic vibrations, and those vibrations are the root cause of both airborne acoustic noise and long-term mechanical stress [1]. They arise when time-varying electromagnetic forces excite structural resonances inside the motor assembly [2]. Radial-Flux Permanent Magnet (RFPM) machines dominate current EV powertrains, but Axial-Flux Permanent Magnet (AFPM) architectures have emerged as high-torque-density alternatives [3], [4]. The two layouts point their primary Maxwell normal forces along different axes, one radially across the air gap and the other axially along the shaft, so each loads the supporting structure in a fundamentally different way [5]. An RFPM machine excites radial ring deformation in the cylindrical stator; an AFPM machine bends its rotor disc into drumhead modes [5]. To separate the contribution of force direction from that of structural compliance, this paper compares the two topologies in a controlled finite-element study that holds mass, pole count, and current density identical.

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A. Related Work

The vibration and dynamic response of radial-flux machines are by now well characterized [1], [6]. Structural excitation in this topology comes from radial Maxwell force harmonics acting across the air gap [7]. The spatial force harmonics have lowest orders set directly by the slot-pole combination, and those low orders excite circumferential deformation modes such as ovalizing or ring modes in the stator lamination core [2], [8].

Axial-flux machines behave differently. Their primary forces act axially along the rotor shaft rather than radially across the air gap, so they bend the rotor disc out of plane into what are commonly called drumhead modes, instead of ovalizing the stator core [5]. Prior work has described these mechanics through general axial force prediction frameworks, slot-pole combination analyses, and targeted dynamic models for yokeless topologies [9]–[11].

What is missing is a controlled comparison. Comparative studies of AFPM and RFPM machines exist for electromagnetic efficiency and power density, but none has isolated the structural vibration behavior of the two topologies under identical constraints [12], [13]. This work supplies that comparison and asks how spatial force direction changes which structural modes are excited. The results show that force orientation sets the specific mode shapes but, on its own, accounts for only a small part of the vibration difference; structural compliance accounts for most of it.

B. Contribution

This work makes three contributions. It establishes a same-basis finite-element framework that compares RFPM and AFPM geometries under matched mass, pole count, and current density, isolating structural dynamic behavior. It then characterizes how the dominant mode shape changes between topologies, showing that radial force waves produce circumferential ring deformation while axial force waves excite out-of-plane rotor disc drumhead modes. Finally, it decomposes the vibration response, shows that structural compliance rather than force magnitude governs the overall vibration magnitude, and maps, by a finite-element speed sweep, where the axial force harmonics cross the disc resonance.

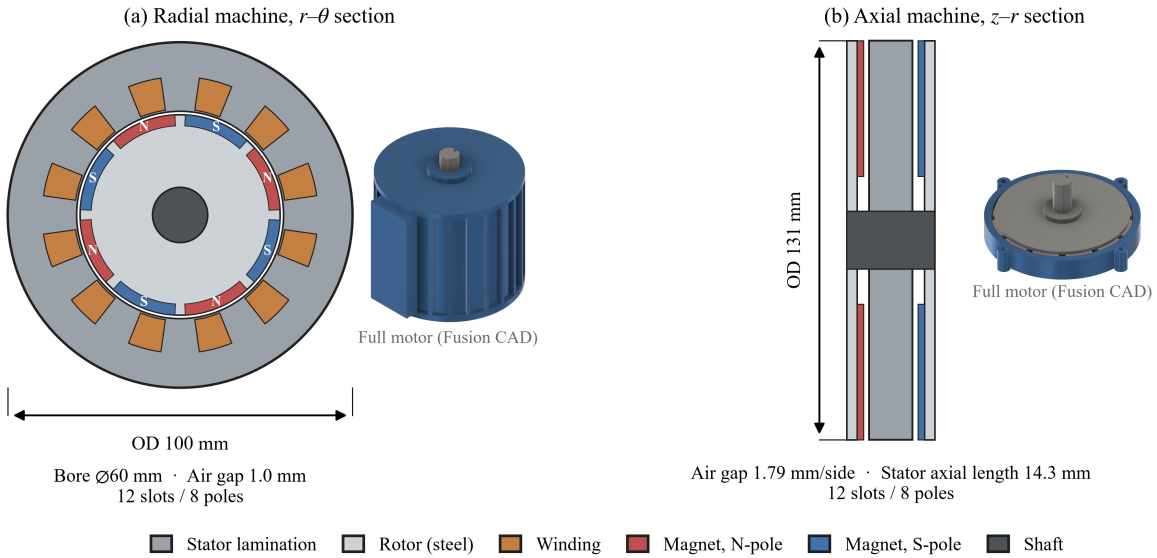


Fig. 1. Cross-sections of the two machines drawn to scale. (a) Radial-flux machine in the radial-tangential (r–θ) plane. (b) Axial-flux machine in the axial-radial (z–r) plane, dual-rotor N-S facing (through-flux TORUS-NS).

C. Scope

The study compares the component on which each machine's Maxwell normal force acts directly and without balance. In the radial machine this is the stator core, which carries the radial force waves into the housing. In the dual-rotor axial machine the central stator is pulled equally by the two rotors, so the net axial force on it cancels to first order, while each rotor disc carries the full one-sided attraction into the shaft and bearings. The rotor disc is therefore the axial counterpart of the radial stator in the force path. Neither component is the final noise radiator, which is the housing in radial machines and the end cover in axial ones [1]. The results describe structural vibration at the first compliant member, not radiated sound; extending them to the housing, the cover, and acoustic radiation is left to future work. Because the comparison pairs a stator against a rotor, only the rotating masses are matched; the radial stator's mass is not, and part of the compliance difference follows from comparing a thick clamped ring with a thin hub-supported disc. A rotor-to-rotor or stator-to-stator comparison would remove this asymmetry and is also left to future work.

II. Motor Topologies and Modeling

To set a controlled baseline for structural evaluation, both machines are designed as 12-slot, 8-pole permanent magnet synchronous motors (PMSMs) with fractional-slot, single-layer concentrated windings (winding factor 0.866), modeled in Autodesk Fusion 360. The two topologies are sized to hold pole count, rotating mass, and current density equal, at 432 A-turns per slot (72 mm^2 at 6 A/mm^2) [14]. Fig. 1 shows the two cross-sections to scale, and Table I summarizes the matched baseline parameters.

A. Radial-Flux Design

The RFPM machine uses an inner-rotor surface-PM layout. Its stator lamination core has an outer diameter of 100 mm, a bore diameter of 60 mm, and an active stack length of 60 mm. Eight surface-mounted magnets sit on the rotor out to an outer radius of 29 mm, separated from the stator bore by a 1.0 mm radial air gap. Each slot carries a total copper cross-sectional area of 72 mm^2.

TABLE I
Key Parameters of the Two Machines

Parameter	Radial	Axial
Configuration	12-slot, 8-pole	12-slot, 8-pole
Topology	inner-rotor SPM	dual-rotor AFPM
Rotor / disc OD	58 mm	131 mm
Stator OD / active	100 mm	48–124 mm dia
Axial length	60 mm stack	14.3 mm tooth
Air gap	1.0 mm	1.79 mm per side
Magnets	8, surface	8 per rotor
Shaft diameter	16 mm	19 mm
Slot copper area	72 mm^2	72 mm^2
Rotating mass	1.320 kg	1.320 kg
Rotating I_{zz}	5.01×10^{-4}	2.26×10^{-3} kg m^2

B. Axial-Flux Design

The AFPM machine is a dual-rotor, single-stator design with N-S facing magnets (through-flux TORUS-NS) and an outer disc diameter of 131 mm, a value chosen to balance the radial-to-axial structural trade-offs reported in single- versus double-rotor evaluations [15], [16]. Each rotor disc carries eight surface-mounted permanent magnets of 2.15 mm thickness over a 3.36 mm back-iron, separated from the central 12-tooth stator by an axial air gap of 1.79 mm per side, with an active stator tooth length of 14.3 mm. The hub outer radius is 19.1 mm and the shaft bore radius is 9.55 mm. The 8-pole count is a controlled variable, matched deliberately to the radial baseline.

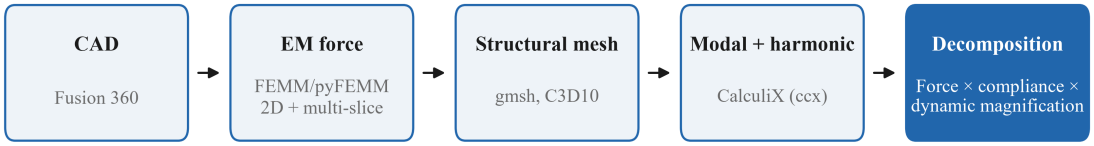


Fig. 2. Analysis pipeline: Fusion 360 CAD, FEMM 2D electromagnetics (radial loaded sweep and axial multi-slice quasi-3D), gmsh structural meshing, CalculiX modal, static, and steady-state dynamic solves, and Python decomposition.

C. Controlled Variables

Baseline equivalence rests on two fixed quantities, a rotating mass of 1.320 kg and an identical conductor current density in both machines. The comparison is therefore made at matched electrical and magnetic loading rather than at matched output; rated torque is not equalized between the machines. Two further quantities are not matched. The axial machine has a polar moment of inertia I_{zz} that is 4.5 times greater than the radial machine (2.26×10^{-3} kg·m^2 versus 5.01×10^{-4} kg·m^2), and this follows unavoidably from redistributing material from a radial to an axial layout rather than from any design choice. The air gap also differs, 1.0 mm for the radial machine against 1.79 mm per side for the axial one, reflecting the different mechanical clearances of a radial bore and a pair of spinning discs. Because air-gap flux density falls as the gap widens, the larger axial gap lowers the axial force, so the force ratio reported here understates rather than overstates the axial penalty. With mass and current density pinned, the structural analysis compares the stationary radial stator ring against the spinning axial rotor disc, which is the compliant element that transmits vibration in each topology.

III. Electromagnetic Force Analysis

A. Modeling Approach

The time-varying electromagnetic forces on both topologies were computed with a 2D magnetostatic finite-element framework built in FEMM and driven through pyFEMM. Fig. 2 shows the full pipeline from CAD through the electromagnetic and structural solvers to the decomposition. Both machines run under rated current over a full electrical period at a rated speed of 3000 r/min, which puts the fundamental electrical frequency at $f_e = 200$ Hz for the 8-pole ($p = 4$ pole-pair) configuration. Following standard sizing formulations [17], the current angle is set to $\gamma = 0°$, consistent with maximum-torque-per-ampere control for surface-mounted PM topologies. Lamination material is modeled as M19 silicon steel and the magnets as NdFeB N40. The excitation of interest is the loaded Maxwell stress under rated current, not cogging torque, since cogging torque is a tangential effect, whereas the ring and drumhead modes studied here are driven by the normal force component [2], [5].

B. Radial-Flux Machine Modeling

The RFPM machine is evaluated with a standard 2D planar cross-sectional model in the radial-tangential (r–θ) plane. Integrating the Maxwell stress tensor along the mid-air-gap contour extracts the time-domain normal radial force density distribution $f_r(\theta, t)$,

$$f_r(\theta,t) = \frac{1}{2\mu_0}\left(B_r^2(\theta,t) - B_\theta^2(\theta,t)\right), \tag{1}$$

where B_r and B_θ are the radial and tangential flux density components and $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space. Integrating $f_r(\theta, t)$ across the tooth pitch angle θ_t and multiplying by the active stack length L_{stk} then gives the total radial force per stator tooth.

C. Axial-Flux Machine Multi-Slice Modeling

Line speed and magnet width both change with radius in the AFPM topology. To capture the resulting axial force variation without paying for a full 3D transient FEA, a quasi-3D multi-slice approach is used. The approximation has been validated for axial-flux evaluations [18], though it carries known fringing and leakage limits [19], [20]. The active radial span between inner radius R_i and outer radius R_o is partitioned into N_{slice} concentric cylindrical rings, and each ring is unrolled into an equivalent 2D planar model in the axial-tangential (z–θ) plane at its mean radius r_k.

Boundary conditions along the radial slice edges enforce magnetic vector potential $A = 0$ Wb/m. At each slice k, the normal axial force density $f_z(r_k, \theta, t)$ follows from Maxwell stress tensor integration,

$$f_z(r_k,\theta,t) = \frac{1}{2\mu_0}\left(B_z^2(r_k,\theta,t) - B_\theta^2(r_k,\theta,t)\right), \tag{2}$$

where B_z and B_θ are the axial and tangential flux density components at slice k. Summing the integrated slice forces across the radial extent gives the total axial force $F_z(t)$ on a rotor disc surface [19],

$$F_z(t) = \sum_{k=1}^{N_{\text{slice}}} f_z(r_k,\theta,t)\cdot 2\pi r_k \cdot \Delta r, \tag{3}$$

where $\Delta r = (R_o - R_i)/N_{\text{slice}}$. Convergence was checked across $N_{\text{slice}} \in \{5, 9, 15\}$, giving net axial forces of 242.2, 242.3, and 242.6 N with a total spread of 0.14%, defined as $(F_{z,\max} - F_{z,\min})/F_{z,\min}$ over the evaluated slice counts.

D. Air-Gap Field and Force Validation

The loaded air-gap fields set the input to every structural solve, so they are checked first. Fig. 3 plots the air-gap flux density and Maxwell stress for both machines at one instant. The radial machine reaches a peak radial flux density of $B_r = 0.51$ T and the axial machine a peak axial flux density of $B_z = 0.76$ T. The corresponding instantaneous Maxwell

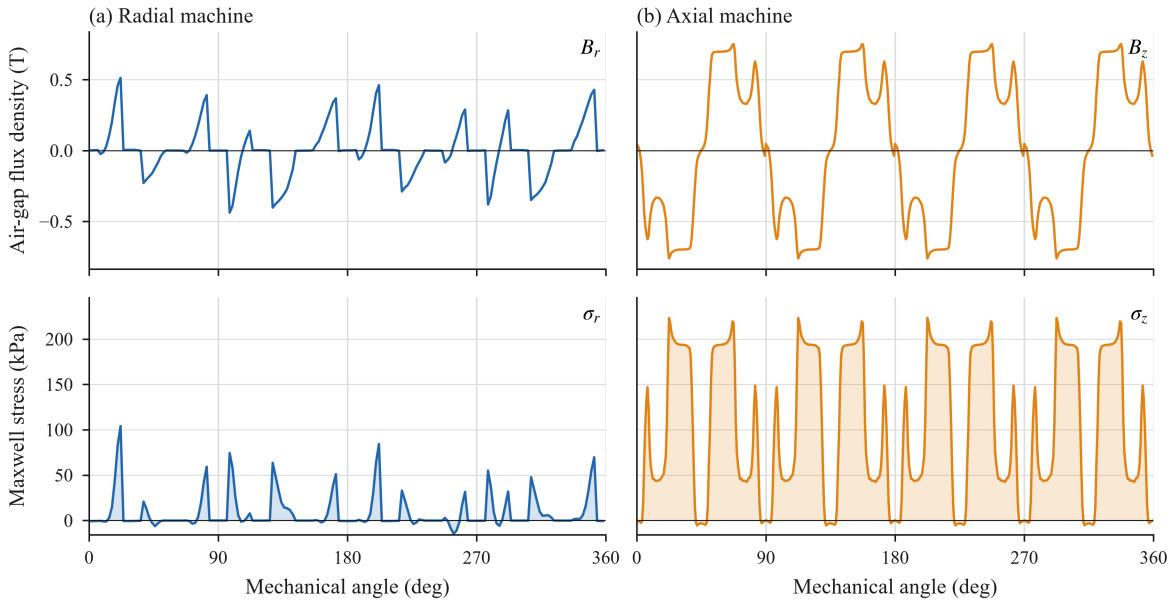


Fig. 3. Loaded air-gap flux density and Maxwell stress at one time step, used to check the force inputs. Axial B_z peaks at 0.76 T. The radial trace was sampled at $r = 36$ mm, inside the stator slots, and is provisional (Section III-D).

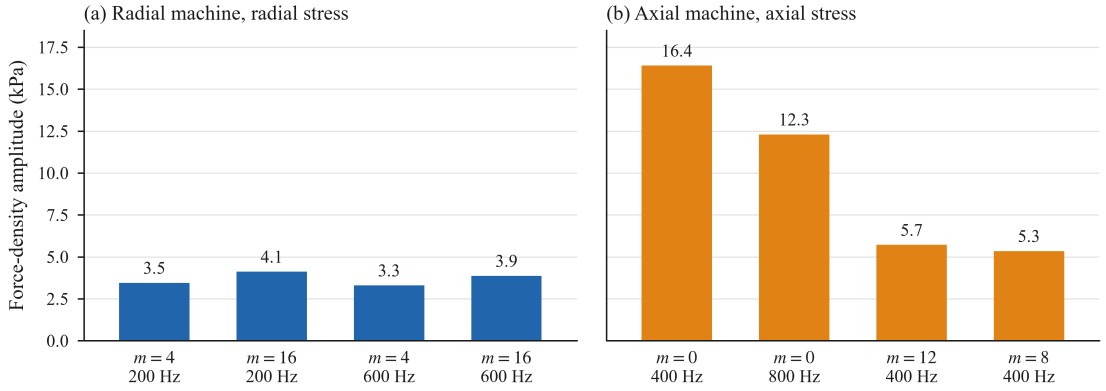


Fig. 4. Vibration-driving force harmonics from the loaded FEMM sweeps. (a) Radial machine, orders $m = 4$ and $m = 16$ at 200 and 600 Hz. (b) Axial machine, dominated by the order-0 breathing force at $2f_e = 400$ Hz.

stress peaks are 104 kPa (radial) and 223 kPa (axial), while the overall peak air-gap pressures are 271 kPa and 232 kPa. The axial values are consistent with the rated loading. The radial values are not reliable. The radial field was sampled along a circle of radius 36 mm, which passes through the stator slots and teeth rather than the 29 to 30 mm air gap, so the Maxwell stress it gives is not the air-gap force on the stator. This explains the flat stretches in the radial trace of Fig. 3. The radial force harmonics in Table II, and every ratio in this paper that depends on them, are therefore provisional until the radial sweep is repeated at the mid-gap radius of 29.5 mm. That rerun will also check the radial harmonic frequencies, which appear at odd multiples of f_e where the B^2 dependence of the force predicts even multiples.

TABLE II
DOMINANT VIBRATION-RELEVANT FORCE HARMONICS (RADIAL VALUES PROVISIONAL)

Machine	Spatial order	Frequency (Hz)	Amplitude (N/m^2)
Radial	4	200	3452
Radial	16	200	4117
Radial	4	600	3298
Radial	16	600	3867
Axial	0	400	16400
Axial	0	800	12300
Axial	12	400	5716
Axial	8	400	5347

E. Force Harmonic Results

To establish the dynamic excitation profiles, the loaded normal Maxwell force distributions under rated current (3000 r/min, 200 Hz base electrical frequency) were decomposed into spatial and temporal harmonics by two-dimensional Fast Fourier Transform (2D FFT). The results appear in Fig. 4 and Table II.

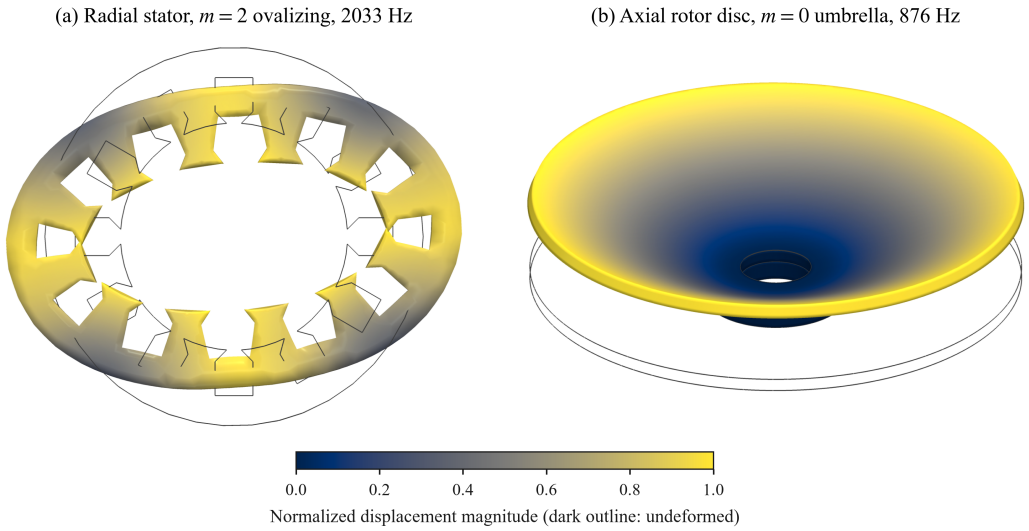


Fig. 5. Contrasting mode shapes, with deformation exaggerated for visibility and normalized to unit amplitude. (a) Stator $m = 2$ ovalizing at 2033 Hz, shown free-free as a conceptual deformation type, since the clamped-rim stator used for the comparison cannot ovalize. (b) Disc $m = 0$ umbrella (drumhead) mode at 876 Hz, which drives the 400 Hz axial response.

In the RFPM topology, normal forces act radially across the air gap. The 12-slot, 8-pole configuration sets the spatial order distribution, with the primary spatial force harmonics at orders $m = 4$ and $m = 16$. In time, the dominant radial excitation lands at 200 Hz (fundamental electrical frequency) and 600 Hz (3rd electrical harmonic).

In the AFPM topology, normal forces act axially along the rotor shaft. Here a spatially uniform order $m = 0$ breathing force dominates the rotor disc, joined by spatial harmonics at orders $m = 8$ and $m = 12$. Because force scales with B^2, all axial harmonics fall at even multiples of f_e, so the primary dynamic axial excitations occur at 400 Hz (2nd electrical harmonic) and 800 Hz (4th electrical harmonic). Static DC components, such as the constant mean axial attraction of about 38.6 kPa, are left out of the dynamic harmonic analysis because they produce static pre-stress rather than vibration.

IV. Structural Modal Analysis

A. Free-Free Modal Setup

Before applying any operational boundary condition, an unconstrained (free-free) modal analysis was run in PrePoMax with the CalculiX solver to verify mesh integrity. It recovered six rigid-body modes with near-zero eigenfrequencies ($f < 10^{-3}$ Hz) and the canonical $m = 2$ ovalizing mode at 2033 Hz, confirming kinematic continuity with no uncoupled nodes. Fig. 5 contrasts this free-free stator ovalizing shape with the axisymmetric disc umbrella mode.

The operating machine is not free-free. In an EV traction motor the stator core is press-fit into an outer aluminum housing, so all operational modal and harmonic evaluations apply fixed boundary conditions ($u_r = u_\theta = u_z = 0$) to the outer cylindrical surface of the stator lamination core.

B. Radial Stator Modes

In the free-free state the canonical stator mode is the $m = 2$ elliptical ovalizing pair at 2033 Hz. Press-fitting the core into the housing changes this. A fully clamped outer rim cannot ovalize, so the lowest clamped mode is not a shifted ovalizing mode but an $m = 12$ tooth-flapping mode at 35,875 Hz, in which the stator teeth flex locally against the fixed yoke. The ring-versus-drumhead contrast of Fig. 5 therefore describes the deformation type each force direction favors, while the clamped stator actually analyzed responds through local tooth flexure.

Either way, the clamped first natural frequency sits far above the excitation. At the fundamental electrical frequency $f_e = 200$ Hz, the frequency ratio is $f/f_n = 200/35{,}875 \approx 0.006$. A ratio this small ($f/f_n \ll 1$) places the clamped RFPM stator deep in the quasi-static, stiffness-dominated regime, where dynamic force amplification does not occur anywhere in the motor operating range. Higher-order clamped modes lie at still higher frequencies and stay comfortably sub-resonant.

C. Axial Rotor and Stator Modes

Modal extraction of the AFPM rotor disc under clamped inner-hub boundary conditions builds on established numerical vibration methods for axial motors [9], and it reveals mode classes with distinct spatial symmetry. The fundamental frequency appears at 765 Hz as a degenerate orthogonal pair, the asymmetric out-of-plane rocking modes ($m = 1$ circumferential nodal line). A second group follows at 876 Hz as a fully axisymmetric out-of-plane flexural bending mode, the umbrella mode ($m = 0$).

Which mode actually responds is decided by spatial coupling between force and shape. The primary 400 Hz axial force harmonic is spatially uniform across the disc surface ($m = 0$), so it does almost no work on the anti-symmetric rocking modes ($m = 1$) and drives them only negligibly. It couples instead to the 876 Hz axisymmetric umbrella mode ($m = 0$). At a frequency ratio of $f/f_n = 400/876 \approx 0.46$ the disc sits in a moderately amplified regime, with a dynamic magnification factor of $M \approx 1.27$. The amplification is moderate at this

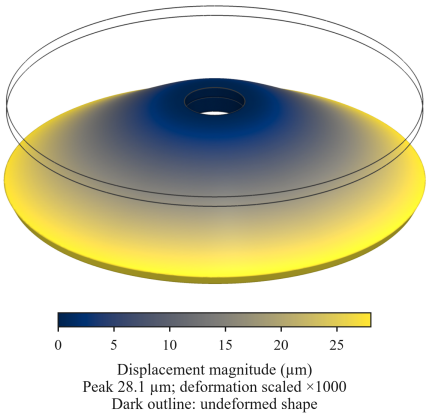


Fig. 6. Static out-of-plane deflection of the AFPM rotor disc under the 400 Hz order-0 axial force, true peak 28 μm shown at $\times$1000 exaggeration.

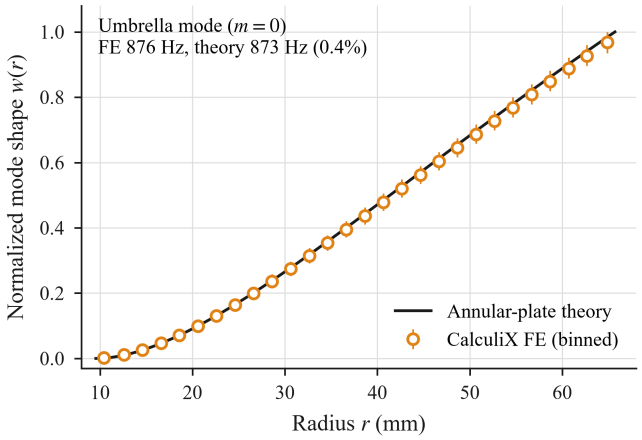


Fig. 7. Disc umbrella mode, finite element versus Kirchhoff annular-plate theory (clamped inner, free outer). The plotted coarse-mesh mode (872.8 Hz) and the fine mesh (875.6 Hz) both agree with theory (872.6 Hz) to within 0.4%, and the shapes overlay.

operating point, but Section V-C shows that the disc does not keep this margin across the speed range. The contrast with the stator at $f/f_{\mathrm{n}} \approx 0.006$ is why the disc reaches a much higher displacement.

V. Harmonic Response and Vibration Prediction

A. Force Mapping and Steady-State Dynamic Method

To predict vibration amplitudes under operational loading, the spatially and temporally resolved Maxwell force densities from Section III were mapped onto the structural finite-element meshes as distributed surface pressure loads. On the RFPM stator core, the combined order $m = 4$ and $m = 16$ radial pressure waves were applied across the inner tooth face surfaces at the fundamental forcing frequency ($f_{\mathrm{e}} = 200$ Hz). On the AFPM rotor disc, the spatially uniform ($m = 0$) pressure was applied across the active axial face at 400 Hz (2nd electrical harmonic).

Harmonic response analysis used a constant structural modal damping ratio of $\zeta = 2.0\%$. For the clamped stator core, operating deep in the sub-resonant regime ($f/f_{\mathrm{n}} \approx 0.006$), inertial effects vanish and a static equilibrium solve reproduces the full steady-state harmonic result exactly. The rotor disc at $f/f_{\mathrm{n}} \approx 0.46$ is a different case, so both static and dynamic steady-state solves were run there to pull out the structural magnification factor on its own.

B. Surface Vibration and Peak Displacement Response

Under load, the two structures respond very differently.

For the clamped RFPM stator core, the combined order $m = 4$ and $m = 16$ radial force harmonics at 200 Hz produce a peak radial surface displacement of 1.38 nm. Just as the sub-resonant frequency ratio predicts, the static and dynamic displacements come out effectively identical, which confirms a stiffness-controlled response with no dynamic overshoot.

The AFPM rotor disc responds far more strongly. Under static application of the 400 Hz order-0 axial force it reaches a peak out-of-plane displacement of 28.06 μm, and under steady-state dynamic excitation at 400 Hz that peak rises to 35.49 μm (Fig. 6). The ratio between them, $M = 35.49/28.06 \approx 1.27$, is the moderate amplification expected as the forcing frequency nears the 876 Hz umbrella mode.

Under matched mass, pole count, and current density, the disc therefore out-deflects the stator core by roughly four orders of magnitude at the baseline supports (35.49 μm against 1.38 nm). Section VI checks these results, and Section VII decomposes the gap into force magnitude, structural compliance, and dynamic magnification.

C. Speed Dependence and Resonance Crossing

The results above hold at one operating point, and the disc's margin from resonance changes with speed. Because the axial order-0 force harmonics sit at even multiples of f_{e}, Fig. 10 tracks both of them against speed. The $2f_{\mathrm{e}}$ harmonic (16.4 kPa) reaches the 876 Hz umbrella mode at 6,570 r/min, and the $4f_{\mathrm{e}}$ harmonic (12.3 kPa) reaches it at 3,285 r/min, both inside a typical traction speed range. Rated speed sits only 9.5% below the second crossing. The two harmonics also cross the 765 Hz rocking pair, at 5,738 and 2,869 r/min, but produce no peak there, because a spatially uniform load cannot excite an antisymmetric mode. The radial stator has no comparable crossing: its first clamped mode at 35,875 Hz is not reached by f_{e} or $3f_{\mathrm{e}}$ below about 1.8×10^5 r/min.

A steady-state finite-element sweep of the bore-clamped disc (fine mesh, $\zeta = 2\%$ on 30 modes) quantifies the effect. At rated speed the $2f_{\mathrm{e}}$ harmonic gives 35.8 μm, matching Section V-B, while the $4f_{\mathrm{e}}$ harmonic at 800 Hz ($f/f_{\mathrm{n}} = 0.91$) gives 126.4 μm, about 3.5 times larger. Their sum, an upper bound that assumes the two peaks coincide, is 162.3 μm. At the crossings the response reaches 532 μm (3,285 r/min) and 709 μm (6,570 r/min). A direct 800 Hz solve (126.43 μm) confirms the sweep (126.46 μm), and a single-degree-of-freedom estimate built from the uniform-load static compliance (1.709 μm/kPa) and the 800 Hz magnification (5.91) gives 124 μm, within 2%. The sweep holds each harmonic at its rated amplitude at every speed, which is a simplification.

The rated-speed disc response is therefore dominated by the 800 Hz harmonic, and the 400 Hz figures used in the

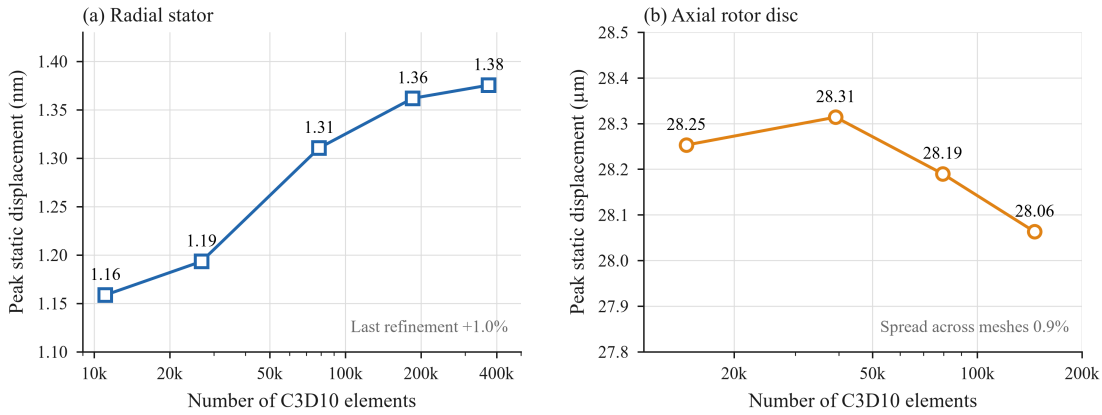


Fig. 8. Mesh convergence of the static displacements. (a) The radial stator approaches a limit near 1.38 nm and sets the ratio precision. (b) The axial disc converges to within 0.9%.

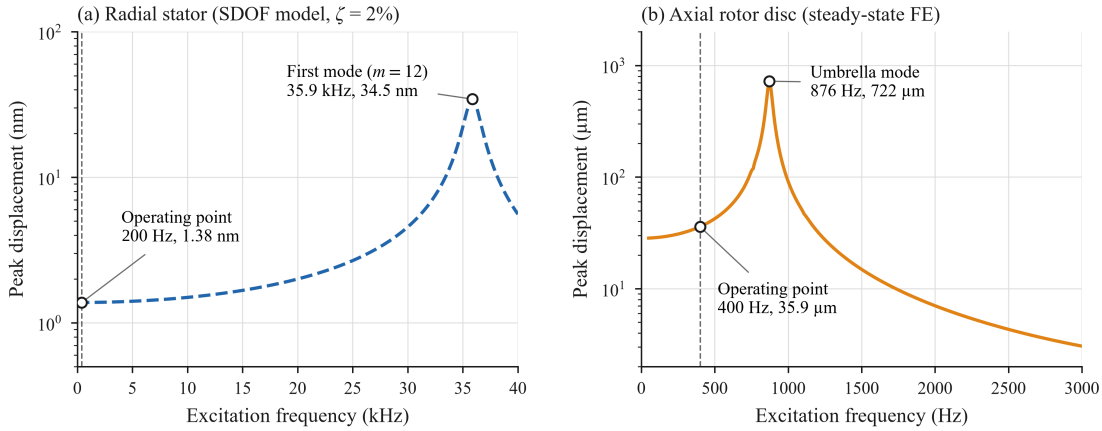


Fig. 9. Frequency response of the two structures. (a) Radial stator, single-degree-of-freedom model ($f_n = 35.9$ kHz, $\zeta = 2\%$) built from the finite-element natural frequency and static displacement; the 200 Hz operating point sits far below resonance. (b) Axial rotor disc, steady-state finite-element sweep (coarse mesh) from 50 to 3000 Hz; a single umbrella resonance at 872 Hz (722 μm) sits well above the 400 Hz operating point (36 μm).

decomposition of Section VII understate it by a factor of about 3.5. Because rated speed lies on the flank of a resonance, small changes in speed, stiffness, or damping move the disc response substantially.

VI. Model Verification and Validation

The comparison rests on two displacement numbers that differ by four orders of magnitude, so each was checked before being trusted. The checks differ in strength. The disc frequency is validated against independent theory, while mesh convergence and the air-gap field check of Section III are verification steps that confirm numerical adequacy rather than physical correctness.

A. Analytical Check on the Disc

The disc umbrella frequency was compared against Kirchhoff annular-plate theory for a clamped-inner, free-outer, axisymmetric plate [21]. The fine-mesh finite-element value of 875.6 Hz and the analytical value of 872.6 Hz agree to 0.34%, the coarse mesh plotted in Fig. 7 gives 872.8 Hz, and the finite-element mode shape overlays the analytical $w(r)$ profile (Fig. 7). This confirms the disc structural model independently of the finite-element solver.

B. Mesh Convergence

Both static displacements were driven to convergence (Fig. 8). The disc static displacement moves from 28.25 to 28.06 μm across 14k to 146k elements, a 0.9% spread, and is treated as converged. The stator static displacement climbs from 1.16 to 1.375 nm across 11k to 372k elements, still rising but slowing to +1.0% at the finest mesh, with a converged limit near 1.38 nm. The stator is the softer link, dominated by local tooth-tip flexure. Aitken extrapolation, assuming the last refinements lie in the asymptotic range, gives a converged limit of about 1.38 nm from the last three meshes and 1.40 nm from the preceding three, so the converged stator displacement lies within roughly 2% above the reported value. The stator result is therefore verified but not independently validated, and the baseline ratio is quoted as an order-of-magnitude figure ($\approx 2.6 \times 10^4$) rather than to full precision.

C. Frequency-Response Cross-Check

A coarse-mesh frequency sweep of the disc from 50 to 3000 Hz shows a single umbrella resonance at 872 Hz with a 722 μm peak, and places the 400 Hz operating point at 36 μm on the sub-resonance flank (Fig. 9). The fine-mesh sweep of

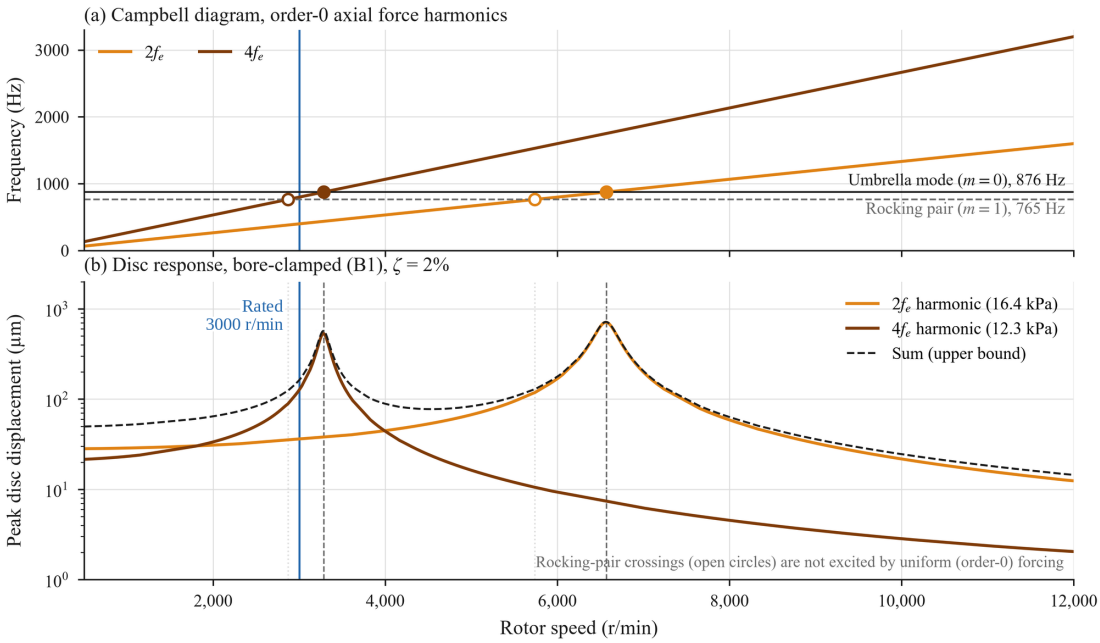


Fig. 10. Disc speed sweep. Top: Campbell diagram of the $2f_e$ and $4f_e$ order-0 harmonics against the 765 Hz rocking and 876 Hz umbrella modes. Bottom: steady-state finite-element disc response to each harmonic, held at its rated amplitude. At the rated 3,000 r/min the 800 Hz harmonic gives 126 μm against 36 μm at 400 Hz.

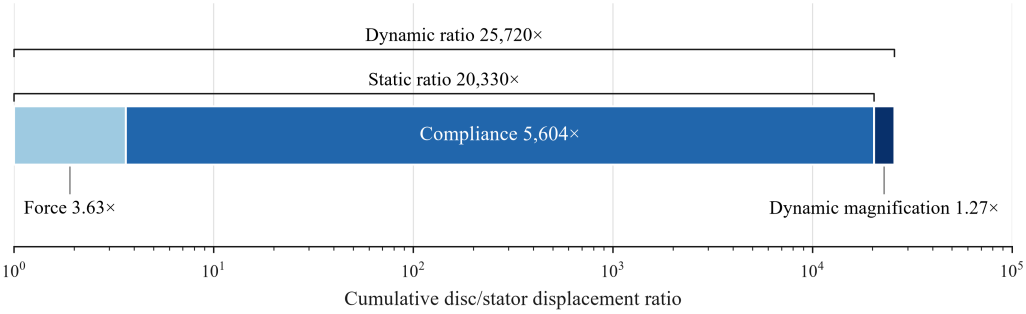


Fig. 11. Two-tier waterfall of the peak displacement gap. Force magnitude (2.17×) and static compliance (9,390×) form the static ratio; the disc dynamic magnification (1.27×) lifts it to the full 25,720×. Radial-dependent values are provisional.

Section V-C peaks at 875.1 Hz. Independent routes therefore agree on the umbrella frequency, the fine-mesh modal solve (875.6 Hz), the analytical plate (872.6 Hz), and the coarse- and fine-mesh frequency responses (872 and 875.1 Hz), and the 36 μm response point matches the steady-state dynamic result of 35.49 μm.

One methodological note applies to both structures. Below the first mode the stator modal steady-state-dynamics solve converges too slowly to be reliable, so a quasi-static solve is used there instead and is exact in this regime (amplification 1.00003). The disc steady-state-dynamics solve is valid and is used directly. Complex displacement magnitude $\sqrt{\mathrm{re}^2 + \mathrm{im}^2}$ is used throughout.

VII. Results and Discussion

A. Quantitative Decomposition of the Vibration Response Gap

Every ratio in this section involves the radial stator and is therefore provisional until the radial force sweep is repeated in the air gap (Section III-D); the axial quantities are not affected. The raw dynamic displacement gap between topologies is $\delta_{\text{axial}}/\delta_{\text{radial}} \approx 25{,}720$. To see where it comes from, the structural response is factored into two independent parts, a static structural response and a dynamic magnification. The static part is then split again, under an applied peak pressure wave convention, to tell force amplitude apart from structural compliance,

$$\frac{\delta_{\text{axial, dyn}}}{\delta_{\text{radial, dyn}}} = \left(\frac{\delta_{\text{axial, stat}}}{\delta_{\text{radial, stat}}}\right)\left(\frac{M_{\text{axial}}}{M_{\text{radial}}}\right) = \left(\frac{F_{\text{axial}}}{F_{\text{radial}}} \cdot \frac{C_{\text{axial}}}{C_{\text{radial}}}\right)\left(\frac{M_{\text{axial}}}{M_{\text{radial}}}\right) \quad (4)$$

where δ_{stat} is the static equilibrium displacement under rated loading, M is the dynamic magnification factor, F is the amplitude of the pressure actually applied to each structure (kPa), and C is the static compliance under that load pattern. On the disc, F is the uniform order-0 amplitude of 16.4 kPa and $C = 1.709$ μm/kPa comes from a separate uniform 1 kPa static solve; their product, 28.03 μm, reproduces the rated

TABLE III
TWO-TIER QUANTITATIVE FACTORIZATION OF THE PEAK DISPLACEMENT GAP (RADIAL VALUES PROVISIONAL)

Decomposition Tier / Metric	**Radial (RFPM)**	**Axial (AFPM)**	**Ratio (Axial / Radial)**
Tier 1: Primary Independent Decomposition			
Static Displacement (δ_{stat})	1.38 nm	28.06 μm	20,330×
Dynamic Magnification (M)	1.00	1.27	1.27×
Total Peak Dynamic Displacement (δ_{dyn})	**1.38** nm	**35.49** μm	**25,720**×
Tier 2: Force and Compliance Sub-Breakdown of Static Response			
Applied Force Density (F)	7.57 kPa	16.40 kPa	2.17×
Static Compliance (C)	0.182 nm/kPa	1.709 μm/kPa	9,390×

TABLE IV
BOUNDARY-CONDITION SENSITIVITY MATRIX AND DYNAMIC DISPLACEMENT RATIOS (PROVISIONAL)

Support Pair	**AFPM Disc (δ_{axial})**	**RFPM Stator (δ_{radial})**	**Ratio ($\delta_{\text{axial}}/\delta_{\text{radial}}$)**	**Dominant Mechanism**
B1 / S1 (Baseline)	35.49 μm	1.38 nm	25,720×	Hub Cantilever vs. Fixed Ring
B2 / S2 (Intermediate)	15.51 μm	2.554 nm	6,073×	Dual-Supported Disc vs. Pad Stator
B3 / S1 (Stiffened Disc)	0.220 μm	1.38 nm	159×	Rim-Clamped Disc vs. Fixed Ring
B3 / S2 (Most Favorable)	0.220 μm	2.554 nm	86×	Fully Stiffened Disc vs. Pad Stator

static displacement to 0.1%. On the stator, F is the in-phase crest sum of the $m = 4$ and $m = 16$ harmonics, an upper bound on the instantaneous pressure, and C is the resulting static displacement per unit of that pressure, so the radial Tier 2 split holds by construction. Only the Tier 1 split into static response and dynamic magnification is fully independent.

Fig. 11 shows the decomposition as a waterfall, and Table III lists the terms. They are as follows.

1) **Static Response Gap (20,330×).** Under static rated force distributions, the AFPM rotor disc reaches $\delta_{\text{axial, stat}} = 28.06$ μm while the clamped RFPM stator core yields $\delta_{\text{radial, stat}} = 1.38$ nm. Their ratio of 20,330× folds together load magnitude and structural boundary conditions, with dynamics set aside.
2) **Dynamic Magnification Gap (1.27×).** The rotor disc umbrella mode, at $f/f_{\text{n}} \approx 0.46$, sees moderate amplification ($M_{\text{axial}} = 1.27$); the deep sub-resonant stator at $f/f_{\text{n}} \approx 0.006$ sees none ($M_{\text{radial}} = 1.00$). Applying the disc magnification to the 20,330× static ratio recovers the full dynamic displacement gap, 25,720× (35.49 μm/1.38 nm).
3) **Sub-Breakdown of Static Response.** Splitting the static ratio into force and compliance shows the uniform axial pressure (16.4 kPa) exceeding the peak radial wave (7.57 kPa, the in-phase crest sum of the $m = 4, 16$ harmonics) by only $F_{\text{axial}}/F_{\text{radial}} = 2.17\times$. The compliance ratio ($C_{\text{axial}} = 1.709$ μm/kPa against $C_{\text{radial}} = 0.182$ nm/kPa) supplies the other 9,390×. The two multiply to $2.17 \times 9{,}390 = 20{,}340\times$, matching the measured static ratio (20,330×) to within 0.1%.

The decomposition is lopsided. Read together, the two tiers show force magnitude contributing a modest 2.17× while structural compliance supplies the remaining 9,390×. That compliance term is not support stiffness alone. Because compliance is normalized by pressure amplitude, it absorbs everything else that differs between the two cases: the support condition, the geometry of a disc against a ring, and the spatial order of the load. The radial load is an $m = 4$, $m = 16$ wave that partly self-cancels around the ring, while the axial order-0 load engages the disc bending fully. The boundary-condition study below isolates the support part of that combination. Prior studies have addressed rotor structural design [22], pole-slot selection [23], and the effect of winding configuration on force and vibration [10]. The decomposition points instead to the rotor structure as the main lever on axial-flux vibration.

B. Boundary-Condition Sensitivity and Penalty Robustness

The baseline supports, inner-hub clamping for the disc (B1) and outer-rim fixing for the stator (S1), are close to the most and least compliant options available, so the baseline ratio of $\approx 2.6 \times 10^4$ may overstate the gap. A boundary-condition sensitivity matrix tests this (Fig. 12). Three rotor disc states (B1: inner-hub clamped, B2: multi-point simply supported at inner and outer radii, B3: outer-rim clamped) are paired against two stator states (S1: outer-rim press-fit fixed, S2: four periodic mounting pads). A spinning disc cannot be supported to ground at its rim, so B2 and B3 are idealizations rather than buildable mounts. They bound what rim stiffening could achieve, for example a spacer ring tying the two rotor discs together outside the stator, which under the symmetric order-0 load acts approximately as a rim constraint. The sensitivity runs used a hub clamp slightly wider than the baseline ($r < 10.5$ mm, with modes at 788 and 892 Hz), and will be repeated with the baseline bore clamp together with the corrected radial loads.

Table IV gives two results. Under the intermediate pairing, a multi-point supported disc (B2) against a pad-mounted stator (S2), the ratio is 6,073× (≈ 6,100×). Under the pairing most favorable to the disc, an outer-rim-clamped disc (B3) against the more flexible pad-mounted stator (S2), the disc still deflects

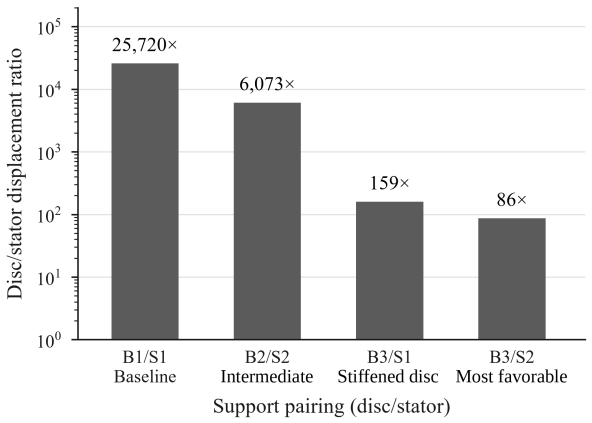


Fig. 12. Disc-to-stator displacement ratio across boundary-condition pairings, from the 25,720× baseline down to a minimum of 86×. The ratio never reaches parity.

0.220 μm against the stator's 2.554 nm, a ratio of 86×. The ratio therefore spans more than two orders of magnitude across supports but does not reach parity in any configuration tested. The direction of the result is robust; its magnitude is set by the boundary conditions, and the intermediate B2/S2 value is the most representative single figure.

Tables III and IV report ratios computed from unrounded solver outputs, which can differ slightly from ratios of the rounded values shown, but the gap should be read as an order of magnitude. The stator displacement carries a mesh uncertainty of roughly 2% (Section VI), while the disc displacement is fully converged. The pad-mounted stator variants (S2) use the study-mesh value of 2.554 nm rather than a reconverged one. Both tables use the 400 Hz harmonic only; Section V-C shows that the 800 Hz harmonic raises the disc response at rated speed to about 3.5 times these figures.

C. Design Implications and Structural Limitations

The design implication follows from the decomposition. Because force magnitude accounts for only 2.17× of the gap, smoothing the electromagnetic force achieves little on its own. Stiffening the rotor support path has far more effect, as the sensitivity study shows: shifting the disc from an inner-hub cantilever (B1) to an idealized outer-rim constraint (B3) cuts the axial displacement from 35.49 μm to 0.220 μm, a 161× reduction with no change to the electromagnetic design or current loading.

Study Limitations. The models are bare steel: they omit magnet mass on the disc, copper windings, slot insulation, and the housing. Added disc mass would lower its natural frequencies and move the resonance crossings of Section V-C to lower speeds, possibly onto rated speed itself. The speed sweep also holds each harmonic at its rated amplitude. Rotating effects on the disc, centrifugal stiffening and gyroscopic splitting of the rocking modes, are neglected. The two machines are modeled at different electromagnetic fidelities, a 2D planar model for the radial machine and a quasi-3D multi-slice model for the axial one, and no full 3D electromagnetic solve confirms the axial forces. Structural forcing uses the dominant harmonic with zero load phase from a magnitude-only FFT and in-phase crest-sum peak pressures, the electromagnetic geometry is estimated, and damping is assumed at 2%. Rated torque is not matched, so the comparison holds at equal current density rather than equal output. Finally, the results are displacements, not radiated sound. The direction of the result rests on the compliance gap ($C_{\text{axial}}/C_{\text{radial}} \sim 10^3$ to 10^4), which these simplifications are unlikely to reverse, but the magnitudes should be read with them in mind.

VIII. Conclusion

This study compares the structural vibration of a radial-flux stator and an axial-flux rotor disc, the first compliant member in each machine's force path, under matched total mass, pole count, and current density.

At rated speed and the 400 Hz axial harmonic, the disc out-deflects the clamped stator by a factor between about 86 and 2.6×10^4 depending on the supports, with an intermediate support pair giving about 6×10^3. The decomposition attributes the gap mainly to structural compliance (about 9,400× at the baseline), with force magnitude ($\sim$ 2.2×) and dynamic amplification ($\sim$ 1.3×) contributing little. These ratios depend on the radial forces and are provisional until the radial sweep is repeated in the air gap. The compliance term combines the support condition, the geometry of a disc against a ring, and the spatial order of the load. The gap narrows with stiffer supports but does not reach parity in any configuration tested.

A finite-element speed sweep shows that the 400 Hz figures understate the disc response. At rated speed the 800 Hz harmonic drives about 126 μm, roughly 3.5 times the 400 Hz response, and rated speed sits only 9.5% below that harmonic's crossing with the umbrella mode at 3,285 r/min.

For design, electromagnetic force smoothing alone cannot close the gap, whereas stiffening the rotor support path reduces it substantially: moving from an inner-hub cantilever toward an outer-rim constraint, approximated in practice by a ring tying the two rotor rims, cuts the disc displacement by more than 160× with no electromagnetic change. Three extensions would strengthen these conclusions: speed-dependent force amplitudes in the speed sweep, comparison at the housing and end-cover level, and prediction of radiated sound.

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